

ETHAN LYLE CRUZ

MACHINE LEARNING | AI ENGINEERING

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Computer Science graduate with a strong interest in Artificial Intelligence (AI), Machine Learning (ML), software development, and UI/UX design. Experienced in building AI-powered applications, deep learning research projects, and user-centered digital systems. Highly organized, with strong time management skills developed through balancing college academics, freelance TikTok/brand deal work, and a healthy lifestyle. Able to solve real-world problems through intelligent, scalable, and intuitive technology, while continuing to grow in data engineering, cloud computing, and security analysis.

EDUCATION

University of Santo Tomas

Bachelor of Science in Computer Science

- *Dean's Lister — 2nd Year, 1st Semester*
- *Dean's Lister — 4th Year, 2nd Semester*

TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, HTML/CSS, React, Laravel
- **AI / Machine Learning:** LLaMA 3.1, RAG, FAISS, SentenceTransformers, CNN, LSTM, NLP, TensorFlow, Image Classification
- **Data & Databases:** SQLite, MySQL, MongoDB, Firebase, NumPy, Pandas, Data Visualization
- **Tools & Platforms:** Streamlit, Groq API, Git, GitHub, Postman, Strapi, Vercel, Supabase
- **Testing & QA:** Debugging, Unit Testing, QA Testing
- **Security & Networking:** Wireshark, Basic Security Analysis
- **Soft Skills:** Time Management, Stakeholder Communication, Project Management

PROJECTS

AI — Powered Personal Gym Trainer

- Built and deployed a full-stack AI fitness coaching app using LLaMA 3.3, RAG, Groq API, and a custom evidence-based fitness knowledge base.
- Implemented FAISS vector search with SentenceTransformers to deliver context-aware training, nutrition, and hypertrophy coaching responses.
- Integrated a Vision-Language Model body fat analyzer using Groq Vision, image upload processing, and structured prompting to estimate body composition from physique images.
- Added user authentication, persistent chat history, and Supabase-backed fitness tracking features.
- Built weight, strength, calorie, and macro tracking dashboards with progress analytics and interactive charts.
- Tech Stack: Python • LLaMA 3.3 • Groq API • RAG • FAISS • SentenceTransformers • Vision-Language Models • Vision Transformers • Streamlit • Supabase • Plotly • GitHub
- Live Demo: coach-e-gym.streamlit.app

EmoTox — Hybrid Toxicity Detection System

- Developed a deep learning toxicity detection system focused on emotion analysis, sarcasm detection, and contextual toxic language.
- Built a hybrid CNN-LSTM emotion classifier and BiLSTM attention-based sarcasm detector for ensemble toxicity classification.
- Achieved 90.22% accuracy and 82.72% F1-score, outperforming traditional lexicon-based toxicity detection approaches.
- Tech Stack: Python • CNN-LSTM • BiLSTM • NLP • Word2Vec • Deep Learning • Attention Mechanisms • JupyterLab

NLP LLM-Configurable GuardRail

- Developed a toxicity detection model using HuggingFace Transformers and PyTorch for real-time harmful text classification in LLM pipelines.
- Trained and evaluated NLP classification models to identify toxic language and support configurable AI moderation systems.

- Integrated the model into a Streamlit interface for deployment as an LLM safety guardrail.
- Tech Stack: Python • HuggingFace Transformers • PyTorch • NLP • Toxicity Detection • Streamlit

AI — Powered Personal Gym Trainer

- Developed a personal portfolio website to showcase projects, technical experience, and academic work in an interactive web interface.
- Integrated an AI chatbot using Groq API to answer questions about my background, projects, skills, and experience.
- Built a FastAPI backend to handle chatbot requests, API routing, and communication between the frontend and the Groq-powered assistant.
- Designed the portfolio to provide recruiters and visitors with a guided, conversational way to explore my work.
- Tech Stack: React • FastAPI • Groq API • Python • JavaScript • REST API • GitHub

EXPERIENCE

WEB DEVELOPER INTERN — RAKSO CT

Jan 2026 - March 2026

- Worked as a lead front-end developer during On-the-Job Training (OJT), supporting web development tasks and user interface improvements.

FREELANCE CONTENT CREATOR & BRAND COLLABORATOR

2023 - Present

- Collaborated with brands on TikTok and Instagram promotional campaigns, managing content creation and audience engagement.
- Balanced freelance work alongside academics, strengthening time management, communication, and self-management skills.

CERTIFICATIONS

- **AWS Foundations: Machine Learning Basics** — AWS Training & Certification, 2026
- **Python for Machine Learning & Data Science Masterclass** — Udemy, 2025
- **Cyber Security 101 Learning Path** — TryHackMe, 2026
- **Legacy Responsive Web Design Certification** — freeCodeCamp, 2023
- **250 Hours On-the-Job Training Certificate** — Rakso CT, 2026